generate and verify a hash value of data. The web3 API provides standard functions for hashing so that the hash computed for a given data and the hash function used for the computation are the same for all the participant applications, resulting in consistent computations. Hashing functions were discussed in chapter 5.

7.1.1 Web3 in Dapp stack

Where does web3 appear in your Dapp stack? Functions supported by web3 can be placed in two categories: those that support the core operations of the blockchain node and those that enable the decentralized application stack on the blockchain. Figure 7.1 illustrates the essential role played by web3 in these two modules, followed by a detailed explanation of each module.

The top part of figure 7.1 shows the *application module,* which has a web server and application code specified in app.js. The bottom part of figure 7.1 is the blockchain client node module, which provides the core blockchain services. Let’s further analyze the layers of the stack in figure 7.1:

- The top layer of the stack is a web client, but it could be any client--mobile or enterprise--that requires the services of the blockchain.
- In the next layer, the web application’s app.js uses the web3.js library to access the blockchain services.
- The layer below is a traditional web server, listening to a port for client requests, and in this case implemented by the Node.js server.
- Web3.js enables app.js application logic to connect to the underlying web3 provider in the blockchain node.